A CONGRUENCE SIEVE SURVIVOR SET WITHOUT LOGARITHMIC DENSITY (ERDŐS PROBLEM #486)

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #486 asks whether for every subset $A \subseteq \mathbb{N}$ and every family of residue classes $X_n \subseteq \mathbb{Z}/n\mathbb{Z}$ ($n \in A$), the survivor set $B = \{m \in \mathbb{N} : \forall\, n \in A \text{ with } n < m,\ m \bmod n \notin X_n\}$ possesses a well-defined logarithmic density. We document the formally verified counterexample constructing a fixed sparse sequence of moduli A and residue sets X_n whose survivor set B has $\liminf_{x\to\infty} \frac{1}{\log x} \sum_{m\in B, m\le x} \frac{1}{m} \le \frac{177}{200} < \frac{49}{50} \le \limsup_{x\to\infty} \frac{1}{\log x} \sum_{m\in B, m\le x} \frac{1}{m}$.

1. Main Theorem

Theorem 1.1 (Formally Verified in Lean 4)**.** *There exist a fixed infinite set $A \subseteq \mathbb{N}_{\ge 2}$ of rapidly growing coprime moduli and residue sets $X_n \subseteq \mathbb{Z}/n\mathbb{Z}$ such that the survivor set*

$$B \;=\; \left\{m \in \mathbb{N}_{\ge 1} : \forall\, n \in A,\ n < m \implies (m \bmod n) \notin X_n\right\}$$

has logarithmic averages $L_B(x) = \frac{1}{\log x} \sum_{1\le m\le x,\, m\in B} \frac{1}{m}$ satisfying

$$\liminf_{x\to\infty} L_B(x) \;\le\; \frac{177}{200} \;<\; \frac{49}{50} \;\le\; \limsup_{x\to\infty} L_B(x),$$

and therefore B does not possess a logarithmic density.

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